Food Hygiene Ratings SQL Project

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1. Perspective and End Consumer

This project looks at **food safety compliance** across the UK, using the Food Standards Agency's food hygiene ratings. It is written for a **local authority environmental health manager**, who needs to know how businesses in their area compare with national figures, which kinds of business and which places are most likely to fall below standard, and how up to date their inspection programme is.

It is a Type 1 project with 40 questions in four groups of ten. The SQL is in `FoodHygiene_Queries.sql` and the database is built by `fsa_setup.sql`.

Section	Questions	What each question does
KPI	Q1–Q10	Produces a headline figure or benchmark
Univariate	Q11–Q20	Describes the distribution of a single variable
Bivariate	Q21–Q30	Shows how one variable changes with another
Multivariate	Q31–Q40	Analyses three or more variables together

Conventions used throughout:

- **Rated** means a business with a Food Hygiene Rating Scheme (FHRS) rating from 0 to 5, used in England, Wales and Northern Ireland. Scotland's Food Hygiene Information Scheme (FHIS) gives Pass or Improvement Required and is covered separately.
- **Broadly compliant** means rated 3 ("generally satisfactory") or above; **below 3** means rated 0, 1 or 2.
- The three **sub-scores** (hygiene, structural, confidence in management) are penalty points, so **lower is better**. Hygiene and structural run from 0 to 25, confidence from 0 to 30.
- **Rating age** is measured to the latest rating date in the file, 30 October 2026, so results do not change with the day the queries are run.
- A **chain** is a business name that appears 50 or more times in the file.

2. Dataset

The data is the FSA's all-authorities open data file, which lists every food business registered with a UK local authority (614,715 rows, latest rating 30 October 2026). It is loaded into the `food_hygiene` database as four tables.

Table	Rows	Contents
`establishments`	614,713	One row per business: rating (as text and as a 0–5 number), rating date, hygiene, structural and confidence sub-scores, postcode, location, new-rating-pending and right-to-reply flags
`business_types`	14	FSA business type and a broader sector (Hospitality, Retail, Public sector and care, Production and supply, Other)
`local_authorities`	363	LA code, name, scheme (FHRS or FHIS) and nation
`rating_descriptions`	6	FSA wording for ratings 0–5, from "Urgent improvement necessary" to "Very good"

Loading

The CSV quotes every field and uses Windows line endings, so `LOAD DATA LOCAL INFILE` reads lines ending `\r\n`. A first attempt using `\n` joined pairs of lines together and loaded only half the rows. Business type and LA names are moved into their lookup tables after loading, so queries join to get them. Nation is derived for each authority: FHIS authorities are Scotland, authorities with BT postcodes are Northern Ireland, the 22 Welsh councils are matched by name, and the rest are England.

Data quality and limitations

Issue	How the project handles it
Two rows are malformed (a stray quote shifts the columns)	Deleted during setup
"Awaiting Inspection" and "AwaitingInspection" both appear	Standardised to one value
FHIS (Scotland) has no 0–5 rating or sub-scores	Covered by its own KPI (Q6) and left out of rating averages

Issue	How the project handles it
Sub-scores are blank for unrated and FHIS businesses	Sub-score questions use rated FHRS businesses only
About 70,000 businesses have no rating date (mostly unrated)	Excluded from rating-age questions
Ratings go back as far as 1998	Kept, and measured as rating age
No region field	Nation is derived; postcode area (letters before the first digit) gives a finer geography

3. KPI Questions (Q1–Q10)

Headline figures and benchmarks that the rest of the analysis is measured against.

Q1

Question: Create a reusable view joining establishments to their business type, sector, local authority and nation. Using it, how many food businesses are listed, and how many are rated, unrated or exempt under each scheme?

Concept: VIEW over a 4-table join + conditional COUNT KPIs

scheme_type	establishments	rated	awaiting_inspection	exempt	awaiting_publication
FHRS	555734	489435	42301	23944	54
FHIS	58979	46331	10359	2289	0

2 rows returned (shown in full).

Interpretation: The file lists 614,713 food businesses. Under FHRS (England, Wales and Northern Ireland), 489,435 of 555,734 have a 0–5 rating, 42,301 are awaiting their first inspection and 23,944 are exempt. Under Scotland's FHIS, 46,331 of 58,979 have a result and 10,359 are awaiting inspection.

Q2

Question: Across all FHRS-rated businesses, what is the average rating, and what share hold the top rating of 5?

Concept: AVG + conditional share

rated_businesses	avg_rating	rated_5	pct_rated_5
489435	4.65	384031	78.5

1 row returned (shown in full).

Interpretation: Rated businesses average 4.65 out of 5, and 384,031 of them (78.5%) hold the top rating of "very good". The scheme is heavily weighted towards high ratings, so the useful comparisons are about the minority below 5.

Q3

Question: What share of rated businesses are broadly compliant (rated 3 or above), and how many are below 3?

Concept: Conditional aggregation KPIs

broadly_compliant	pct_broadly_compliant	below_3	pct_below_3
474559	97.0	14876	3.0

1 row returned (shown in full).

Interpretation: 97.0% of rated businesses are broadly compliant (rated 3 or above). The other 3.0%, 14,876 businesses, are rated 0–2 and need improvement, which is the core caseload for enforcement and revisits.

Q4

Question: How many businesses are rated 0 ("urgent improvement necessary"), and which business types account for most of them?

Concept: Filter + GROUP BY + share of total (window SUM OVER)

business_type	rated_0	pct_of_all_0s
Retailers - other	294	32.3
Restaurant/Cafe/Canteen	282	31.0
Takeaway/sandwich shop	193	21.2
Pub/bar/nightclub	47	5.2
Other catering premises	36	4.0

business_type	rated_0	pct_of_all_0s
Mobile caterer	13	1.4
Manufacturers/packers	13	1.4
Retailers - supermarkets/hypermarkets	12	1.3
Hotel/bed & breakfast/guest house	9	1.0
Hospitals/Childcare/Caring Premises	6	0.7
Distributors/Transporters	4	0.4
Importers/Exporters	2	0.2

12 rows returned (shown in full).

Interpretation: 911 businesses are rated 0, "urgent improvement necessary". Three types account for 84.5% of them: other retailers (32.3%), restaurants and cafés (31.0%) and takeaways (21.2%).

Q5

Question: How large is the inspection backlog: how many FHRS businesses are awaiting their first inspection, as a share of all FHRS businesses?

Concept: Conditional COUNT + share KPI

fhrs_businesses	awaiting_inspection	pct_awaiting
555734	42301	7.6

1 row returned (shown in full).

Interpretation: 42,301 FHRS businesses, 7.6% of the total, are awaiting their first inspection. These are mostly new or newly registered businesses, and the backlog is a key workload measure for an environmental health team.

Q6

Question: Under Scotland's FHIS scheme, what share of inspected businesses pass, and how many need improvement?

Concept: Conditional aggregation on text categories

pass	of_which_eat_safe	improvement_required	pct_pass
42866	477	3465	92.5

1 row returned (shown in full).

Interpretation: In Scotland, 92.5% of inspected businesses pass, including 477 with the higher "Eat Safe" award, and 3,465 require improvement. FHIS uses a different standard from FHRS, so this is not directly comparable with the 97.0% broadly compliant figure in Q3.

Q7

Question: How up to date are ratings? What is the average age of an FHRS rating, and what share are more than two and more than five years old?

Concept: Date arithmetic (DATEDIFF) + conditional shares

avg_age_years	pct_over_2_years	pct_over_5_years
1.8	25.0	7.0

1 row returned (shown in full).

Interpretation: The average FHRS rating is 1.8 years old. A quarter (25.0%) are more than two years old and 7.0% more than five, so a sizeable minority of ratings may no longer reflect current conditions.

Q8

Question: What are the average hygiene, structural and confidence-in-management penalty scores, and what share of rated businesses scored a perfect 0 on all three?

Concept: Several AVG KPIs + compound condition

avg_hygiene	avg_structural	avg_confidence	avg_total	pct_perfect
2.90	3.76	3.77	10.44	23.8

1 row returned (shown in full).

Interpretation: The average penalty scores are 2.90 for hygiene, 3.76 for structural and 3.77 for confidence in management, a total of 10.44. Lower is better, so structure and management are slightly weaker areas than hygiene. 23.8% of rated businesses scored a

perfect 0 on all three.

Q9

Question: How many ratings were issued in the 12 months before the reference date, and what share of those were rated 5 and below 3?

Concept: Date window filter + KPIs

ratings_last_12m	avg_per_month	pct_rated_5	pct_below_3
207828	17319	73.5	5.0

1 row returned (shown in full).

Interpretation: 207,828 ratings were issued in the 12 months to 30 October 2026, about 17,300 a month. Recent ratings are lower than the overall stock: 73.5% were rated 5 and 5.0% below 3, against 78.5% and 3.0% overall. Lower-rated businesses are inspected more often under risk-based scheduling, so they are over-represented in recent inspections.

Q10

Question: Create a reusable view summarising each local authority. Using it, how many FHRS authorities have at least 95% of rated businesses broadly compliant, and how many have more than 5% below 3?

Concept: VIEW (LA-level aggregation) + conditional KPIs

fhrs_authorities	las_95pct_compliant	pct_las_95pct_compliant	las_over_5pct_below_3
329	289	87.8	40

1 row returned (shown in full).

Interpretation: Of 329 FHRS local authorities with 100+ rated businesses, 289 (87.8%) have at least 95% broadly compliant. The other 40 have more than 5% of businesses below 3, and they are the authorities to look at first.

4. Univariate Questions (Q11–Q20)

Each question describes the spread of a single variable.

Q11

Question: How are FHRS ratings distributed from 0 to 5, with the FSA's description of each?

Concept: Frequency table with a join to a lookup + share

rating_num	description	businesses	pct_of_rated
5	Very good	384031	78.5
4	Good	63812	13.0
3	Generally satisfactory	26716	5.5
2	Improvement necessary	7324	1.5
1	Major improvement necessary	6641	1.4
0	Urgent improvement necessary	911	0.2

6 rows returned (shown in full).

Interpretation: Ratings are heavily skewed towards the top: 78.5% are 5 and 13.0% are 4. Only 3.0% fall below 3, and just 911 businesses (0.2%) are rated 0.

Q12

Question: What does the full list of rating values look like, including the non-numeric statuses, across both schemes?

Concept: Frequency table over a text column + share

scheme_type	rating_value	businesses	pct_of_all
FHRS	5	384031	62.5
FHRS	4	63812	10.4
FHIS	Pass	42389	6.9
FHRS	AwaitingInspection	42301	6.9
FHRS	3	26716	4.3
FHRS	Exempt	23944	3.9
FHIS	AwaitingInspection	10359	1.7
FHRS	2	7324	1.2

scheme_type	rating_value	businesses	pct_of_all
FHRS	1	6641	1.1
FHIS	Improvement Required	3465	0.6
FHIS	Exempt	2289	0.4
FHRS	0	911	0.1
FHIS	Pass and Eat Safe	477	0.1
FHRS	AwaitingPublication	54	0.0

14 rows returned (shown in full).

Interpretation: Across both schemes, an FHRS rating of 5 is by far the most common status (62.5% of all listings). The non-rated statuses together, awaiting inspection (8.6%), exempt (4.3%) and awaiting publication, cover about one listing in eight.

Q13

Question: How are hygiene penalty scores distributed among rated businesses?

Concept: Frequency of a discrete numeric variable

hygiene	businesses	pct
0	254465	52.6
5	185743	38.4
10	37284	7.7
15	5769	1.2
20	554	0.1
25	148	0.0

6 rows returned (shown in full).

Interpretation: More than half of rated businesses (52.6%) scored 0 penalty points for hygiene and 91.0% scored 5 or fewer. Only 1.3% scored 15 or more, the level that usually means a serious hygiene problem.

Q14

Question: How are structural penalty scores distributed among rated businesses?

Concept: Frequency of a discrete numeric variable

structural	businesses	pct
0	192808	39.8
5	228617	47.2
10	53120	11.0
15	8354	1.7
20	852	0.2
25	212	0.0

6 rows returned (shown in full).

Interpretation: Structural scores are perfect less often than hygiene: 39.8% scored 0 and the most common score is 5 (47.2%). 1.9% scored 15 or more, so building condition and cleanliness of the premises are a slightly more common issue than food handling.

Q15

Question: How are confidence-in-management penalty scores distributed among rated businesses?

Concept: Frequency of a discrete numeric variable + running (cumulative) share

confidence	businesses	pct	cumulative_pct
0	203720	42.1	42.1
5	210656	43.5	85.6
10	62329	12.9	98.5
20	6833	1.4	99.9
30	425	0.1	100.0

5 rows returned (shown in full).

Interpretation: 42.1% of businesses scored 0 for confidence in management and 85.6% scored 5 or fewer. The top two penalty levels (20 and 30), which indicate poor food safety management, apply to 1.5% of businesses.

Q16

Question: Which types of food business are most common?

Concept: Frequency table (join to lookup) + share

business_type	sector	businesses	pct
Restaurant/Cafe/Canteen	Hospitality	141325	23.0
Retailers - other	Retail	120984	19.7
Other catering premises	Other	76873	12.5
Takeaway/sandwich shop	Hospitality	62925	10.2
Pub/bar/nightclub	Hospitality	50522	8.2
Hospitals/Childcare/Caring Premises	Public sector and care	43388	7.1
School/college/university	Public sector and care	36007	5.9
Mobile caterer	Hospitality	31310	5.1
Retailers - supermarkets/hypermarkets	Retail	15997	2.6
Hotel/bed & breakfast/guest house	Hospitality	14429	2.3
Manufacturers/packers	Production and supply	12234	2.0
Distributors/Transporters	Production and supply	4765	0.8
Farmers/growers	Production and supply	3009	0.5
Importers/Exporters	Production and supply	945	0.2

14 rows returned (shown in full).

Interpretation: Restaurants and cafés are the most common food business (23.0%), followed by other retailers (19.7%), other catering premises (12.5%) and takeaways (10.2%). Hospitality types together make up about half of all listings, while production and supply businesses are just 3.5%.

Q17

Question: In which year was each current FHRS rating issued?

Concept: CTE + YEAR() + share and running total (window SUM)

rating_year	ratings	pct	pct_this_year_or_newer
2026	175357	35.8	35.8
2025	174467	35.6	71.5
2024	66816	13.7	85.1
2023	24610	5.0	90.2
2022	12741	2.6	92.8
2021	6635	1.4	94.1
2020	4052	0.8	94.9
2019	7282	1.5	96.4
2018	5227	1.1	97.5
2017	3398	0.7	98.2
2016	2209	0.5	98.6
2015	1434	0.3	98.9
2014	1249	0.3	99.2
2013	899	0.2	99.4
2012	739	0.2	99.5
2011	606	0.1	99.7
2010	607	0.1	99.8
2009	550	0.1	99.9
2008	183	0.0	99.9
2007	109	0.0	99.9
2006	62	0.0	100.0
2005	48	0.0	100.0
2004	40	0.0	100.0
2003	39	0.0	100.0
2002	19	0.0	100.0
2001	27	0.0	100.0

rating_year	ratings	pct	pct_this_year_or_newer
2000	8	0.0	100.0
1999	12	0.0	100.0
1998	7	0.0	100.0

29 rows returned (shown in full).

Interpretation: 71.5% of current ratings were issued in 2025 or 2026, and 85.1% since the start of 2024. A long tail remains: 1.4% date from before 2016, and the oldest from 1998, which suggests some low-risk businesses have not been re-inspected for many years.

Q18

Question: How many food businesses does each local authority cover? Show the summary statistics and quartiles.

Concept: CTE + window functions (ROW_NUMBER, COUNT OVER) for percentiles + summary statistics

local_authorities	smallest	lower_quartile	median	upper_quartile	largest	mean
363	3	919	1317	2038	10328	1693

1 row returned (shown in full).

Interpretation: The 363 local authorities in the file cover a median of 1,317 food businesses each, with the middle half between 919 and 2,038. The largest has 10,328. The smallest has only 3, probably a special authority such as a port health authority rather than a council.

Q19

Question: How is the total penalty score (hygiene + structural + confidence) distributed? Group rated businesses into score bands.

Concept: CTE + CASE banding + share

score_band	businesses	pct
1. 0 (perfect)	115304	23.8
2. 1-15	285408	59.0
3. 16-30	71550	14.8
4. 31-45	9049	1.9

score_band	businesses	pct
5. Over 45	2652	0.5

5 rows returned (shown in full).

Interpretation: 23.8% of rated businesses have a perfect total of 0 and a further 59.0% score between 1 and 15. Only 2.4% score above 30, which is roughly where ratings fall below 3.

Q20

Question: Which 15 postcode areas have the most food businesses?

Concept: Derived column (REGEXP_SUBSTR) + GROUP BY + LIMIT

postcode_area	businesses	pct_of_uk
B	14661	2.9
BT	14616	2.9
M	10511	2.1
G	10340	2.0
S	9999	2.0
NE	9125	1.8
E	8851	1.7
SE	8416	1.7
NG	8331	1.6
CF	8089	1.6
EH	7749	1.5
W	7727	1.5
SW	7713	1.5
LE	7622	1.5
SA	7374	1.4

15 rows returned (shown in full).

Interpretation: Birmingham (B) and Belfast (BT) are the largest postcode areas, each with 2.9% of food businesses; BT covers the whole of Northern Ireland. Manchester, Glasgow and Sheffield follow at about 2% each, and the top 15 areas together hold just over a quarter of all listings (27.7%).

5. Bivariate Questions (Q21–Q30)

Each question looks at how one variable changes with another.

Q21

Question: How do average ratings and the share below 3 vary by business type?

Concept: Numeric by category: GROUP BY with AVG + share

business_type	rated	avg_rating	pct_below_3
Takeaway/sandwich shop	52159	4.34	5.7
Retailers - other	88748	4.51	5.2
Restaurant/Cafe/Canteen	119630	4.60	3.6
Pub/bar/nightclub	43962	4.67	2.3
Importers/Exporters	411	4.74	3.2
Distributors/Transporters	2612	4.76	2.1
Manufacturers/packers	7792	4.76	1.9
Hotel/bed & breakfast/guest house	10810	4.79	1.4
Retailers - supermarkets/hypermarkets	14259	4.79	2.2
Mobile caterer	24046	4.83	1.4
Hospitals/Childcare/Caring Premises	36018	4.84	0.7
Other catering premises	56856	4.85	1.0
Farmers/growers	1242	4.89	0.2
School/college/university	30890	4.90	0.4

14 rows returned (shown in full).

Interpretation: Takeaways have the lowest average rating (4.34) and the highest share below 3 (5.7%), followed by other retailers (5.2%) and restaurants (3.6%). Schools (0.4% below 3), farmers and growers (0.2%) and care premises (0.7%) have the fewest problems.

Q22

Question: How does the share of businesses rated 5 compare across the broader sectors?

Concept: Numeric by category

sector	rated	pct_rated_5	avg_rating
Public sector and care	66908	90.0	4.87
Other	56856	89.2	4.85
Production and supply	12057	85.3	4.78
Hospitality	250607	74.6	4.59
Retail	103007	73.7	4.55

5 rows returned (shown in full).

Interpretation: Public sector and care premises have the highest share rated 5 (90.0%), while hospitality (74.6%) and retail (73.7%) are about 15 points lower. These two sectors hold most of the rated businesses, so they drive most of the enforcement workload.

Q23

Question: How do ratings compare between England, Wales and Northern Ireland?

Concept: Numeric by category (nation)

nation	rated	avg_rating	pct_rated_5	pct_below_3
Northern Ireland	15506	4.77	83.6	1.2
England	443397	4.65	78.5	3.1
Wales	30532	4.64	76.0	2.5

3 rows returned (shown in full).

Interpretation: Northern Ireland has the best results: an average of 4.77, 83.6% rated 5 and only 1.2% below 3. England (4.65) and Wales (4.64) are close on average, but England has a higher share below 3 (3.1% against 2.5%), while Wales has fewer businesses at the top rating.

Q24

Question: Are older ratings lower or higher? Compare the average rating across bands of rating age.

Concept: CTE + CASE banding of one variable + AVG of another

age_band	rated	avg_rating	pct_rated_5
1. Under 1 year	207828	4.53	73.5
2. 1-2 years	159160	4.66	78.0
3. 2-3 years	55409	4.84	87.7
4. 3-5 years	33011	4.86	88.7
5. Over 5 years	34024	4.84	85.6

5 rows returned (shown in full).

Interpretation: Ratings issued in the last year are lower (4.53 average, 73.5% rated 5) than those two or more years old (about 4.85, 86–89% rated 5). This is not because businesses improve over time: under risk-based inspection, low-rated businesses are revisited often, so their ratings stay recent, while high-rated, low-risk businesses go longer between visits.

Q25

Question: How closely are a business's hygiene and structural scores linked? Calculate the correlation.

Concept: Aggregate formula for Pearson correlation

businesses	correlation_hygiene_structural
483963	0.55

1 row returned (shown in full).

Interpretation: Hygiene and structural scores have a correlation of 0.55, a moderate positive link: businesses with poor food handling also tend to have poorly maintained premises, though each score still carries its own information.

Q26

Question: For each rating from 0 to 5, what is the typical confidence-in-management score?

Concept: GROUP BY one variable + AVG, MIN, MAX of another

rating_num	rated	avg_confidence	min_confidence	max_confidence
5	380274	2.4	0	5
4	62613	7.1	0	10
3	26312	9.6	5	10
2	7264	9.5	0	10
1	6604	19.6	0	20
0	896	24.6	0	30

6 rows returned (shown in full).

Interpretation: The confidence-in-management score rises sharply as the rating falls, from an average of 2.4 for a 5 to 19.6 for a 1 and 24.6 for a 0. Businesses rated 2 and 3 have almost the same confidence score (about 9.5), so a 2 is usually caused by a high hygiene or structural score rather than management.

Q27

Question: Do businesses that have published a right to reply have lower ratings than those that have not?

Concept: Numeric by a flag (two groups)

right_to_reply	rated	avg_rating	pct_below_3
Right to reply published	74	2.28	58.1
No right to reply	489361	4.65	3.0

2 rows returned (shown in full).

Interpretation: Only 74 rated businesses have published a right to reply, and they average just 2.28, with 58.1% below 3. Businesses mainly use the right to reply to respond to a low rating, so this is an indicator of a disputed poor rating rather than a cause.

Q28

Question: Does the month of inspection make a difference? Show the share rated 5 by calendar month for ratings issued in the last three years.

Concept: GROUP BY on MONTH() / MONTHNAME() + share

month_no	month_name	rated	pct_rated_5
1	January	34764	79.0
2	February	37071	78.6
3	March	42854	77.5
4	April	29651	76.7
5	May	33834	76.4
6	June	38567	75.8
7	July	39503	74.7
8	August	33505	72.7
9	September	44289	75.2
10	October	35292	82.3
11	November	31479	79.7
12	December	21588	77.5

12 rows returned (shown in full).

Interpretation: The share rated 5 dips through the summer to a low of 72.7% in August, the busiest season for many food businesses. October's high of 82.3% is inflated by the most recent inspections, where lower ratings are still in their appeal period and not yet published (see Q39).

Q29

Question: Is there a north-south pattern? Compare average ratings across bands of latitude.

Concept: CTE + FLOOR banding of a numeric variable + AVG

latitude_band	rated	avg_rating	pct_below_3
56-57 N	3	4.33	0.0
55-56 N	4787	4.80	1.2
54-55 N	32189	4.71	2.0
53-54 N	89120	4.57	3.9
52-53 N	71075	4.59	3.9
51-52 N	144067	4.54	4.1
50-51 N	33498	4.72	2.3
49-50 N	121	4.92	0.0

8 rows returned (shown in full).

Interpretation: Businesses between 51° and 54° north, which includes London, the Midlands and the large northern cities, have the lowest averages (4.54–4.59) and about 4% below 3. Those further north (54–56°) and on the south coast (50–51°) do better. The 56–57° and 49–50° bands hold too few FHRS businesses to compare, because Scotland uses FHIS. Only the three-quarters of rated businesses with coordinates are included.

Q30

Question: Do chain businesses (names appearing 50+ times) score better than independents?

Concept: CTE + window COUNT OVER (PARTITION BY name) + group comparison

ownership	rated	avg_rating	pct_rated_5	pct_below_3
Chain (50+ outlets)	38015	4.82	87.5	1.4
Independent	451420	4.64	77.7	3.2

2 rows returned (shown in full).

Interpretation: Chain outlets average 4.82 with 87.5% rated 5, against 4.64 and 77.7% for independents. Independents are more than twice as likely to be below 3 (3.2% against 1.4%), which likely reflects the central food safety systems and training that chains provide.

6. Multivariate Questions (Q31–Q40)

Each question analyses three or more variables together.

Q31

Question: How do ratings vary by sector within each nation?

Concept: Two-way GROUP BY (nation x sector) + several measures

nation	sector	rated	avg_rating	pct_rated_5	pct_below_3
England	Public sector and care	59527	4.87	90.2	0.6
England	Other	51928	4.84	89.0	1.0
England	Production and supply	9632	4.78	85.6	1.9
England	Hospitality	228361	4.59	74.8	3.6
England	Retail	93949	4.54	73.4	5.0
Northern Ireland	Other	1795	4.95	96.0	0.4
Northern Ireland	Public sector and care	2713	4.93	94.2	0.1
Northern Ireland	Production and supply	648	4.77	83.0	0.9
Northern Ireland	Retail	3047	4.75	81.4	1.3
Northern Ireland	Hospitality	7303	4.67	77.7	1.9
Wales	Other	3133	4.84	88.4	1.0
Wales	Public sector and care	4668	4.82	85.6	0.7
Wales	Production and supply	1777	4.79	84.4	1.2
Wales	Retail	6011	4.58	73.8	4.0
Wales	Hospitality	14943	4.54	70.3	2.9

15 rows returned (shown in full).

Interpretation: In every nation, hospitality and retail are the weakest sectors and public sector and care the strongest. English retail has the highest share below 3 (5.0%), and Welsh hospitality the lowest share rated 5 (70.3%). Northern Ireland outperforms England and Wales in every sector.

Q32

Question: For the main hospitality business types, how does the share rated 5 change as ratings get older?

Concept: CASE banding + cross-tab with conditional aggregation

business_type	rated	pct5_under_1yr	pct5_1_to_3yrs	pct5_over_3yrs
Takeaway/sandwich shop	52159	54.6	67.1	83.5
Restaurant/Cafe/Canteen	119628	70.1	79.7	87.8
Pub/bar/nightclub	43962	72.8	79.4	80.6
Hotel/bed & breakfast/guest house	10810	80.0	86.7	89.8
Mobile caterer	24046	85.8	89.2	93.2

5 rows returned (shown in full).

Interpretation: For every hospitality type, the share rated 5 is lowest among ratings issued in the last year and highest among ratings over three years old. The gap is widest for takeaways (54.6% against 83.5%), which again shows that weaker businesses are inspected more often. Only about half of takeaways inspected in the last year achieved a 5.

Q33

Question: Using the LA summary, which are the five best and five worst local authorities in each nation by share of businesses below 3 (200+ rated businesses)?

Concept: Uses v_la_summary + CTE + window functions (RANK partitioned by nation, both ways)

nation	grp	la_name	rated	pct_below_3	pct_rated_5	avg_rating
England	Best 5	Stockton On Tees	1453	0.0	95.8	4.94
England	Best 5	Bassetlaw	1003	0.2	97.7	4.96
England	Best 5	Cheltenham	1017	0.3	89.9	4.88
England	Best 5	Cotswold	1027	0.3	94.1	4.93
England	Best 5	West Oxfordshire	1016	0.3	94.9	4.93
England	Best 5	Dorset	4056	0.3	95.4	4.94
England	Best 5	North Kesteven	908	0.3	93.2	4.91
England	Worst 5	Birmingham	8606	10.5	63.7	4.26
England	Worst 5	Blackburn	1146	10.6	68.0	4.32
England	Worst 5	Waltham Forest	1860	12.4	50.7	3.98

nation	grp	la_name	rated	pct_below_3	pct_rated_5	avg_rating
England	Worst 5	Ealing	3207	13.1	57.2	4.15
England	Worst 5	Newham	2453	13.9	48.5	3.95
Northern Ireland	Best 5	Mid and East Antrim	1023	0.2	93.0	4.91
Northern Ireland	Best 5	Antrim and Newtownabbey	986	0.6	87.1	4.82
Northern Ireland	Best 5	Derry City and Strabane	1229	0.7	76.1	4.70
Northern Ireland	Best 5	Ards and North Down	1159	0.8	82.7	4.78
Northern Ireland	Best 5	Lisburn and Castlereagh City	861	0.8	91.6	4.89
Northern Ireland	Worst 5	Mid Ulster	1300	1.3	88.9	4.85
Northern Ireland	Worst 5	Causeway Coast and Glens	1351	1.4	87.6	4.82
Northern Ireland	Worst 5	Newry, Mourne and Down	1599	1.4	88.6	4.83
Northern Ireland	Worst 5	Fermanagh and Omagh	1332	1.4	85.4	4.79
Northern Ireland	Worst 5	Belfast City	2907	2.3	72.6	4.59
Wales	Best 5	Carmarthenshire	1889	0.7	83.4	4.77
Wales	Best 5	Wrexham	1184	0.9	94.7	4.91
Wales	Best 5	Gwynedd	1973	1.2	90.4	4.85
Wales	Best 5	Anglesey	773	1.4	90.6	4.85
Wales	Best 5	Monmouthshire	992	1.4	78.6	4.68
Wales	Worst 5	Torfaen	727	3.3	71.1	4.55
Wales	Worst 5	Cardiff	3031	3.4	70.4	4.53
Wales	Worst 5	Caerphilly	1385	4.5	63.0	4.42
Wales	Worst 5	Neath Port Talbot	1150	5.6	67.3	4.45
Wales	Worst 5	Blaenau Gwent	591	5.6	50.3	4.24

32 rows returned (shown in full).

Interpretation: England has the widest spread: the best authorities, such as Stockton-on-Tees and Bassetlaw, have almost no businesses below 3, while Newham (13.9%), Ealing (13.1%) and Waltham Forest (12.4%) are the worst. Seven authorities appear in England's best group because of a tie at 0.3%. Even Northern Ireland's worst authority, Belfast (2.3%), is better than the UK average, while Wales ranges from Carmarthenshire (0.7%) to Blaenau Gwent and Neath Port Talbot (5.6%).

Q34

Question: Using the LA summary, which authorities combine a high share of businesses below 3, a large inspection backlog and old ratings? Show those above the national figure on all three (200+ rated).

Concept: Uses v_la_summary + CROSS JOIN to national averages + multi-condition filter

la_name	nation	rated	pct_below_3	national_pct_below_3	pct_awaiting	national_pct_awaiting	avg_rating_age	national_avg_age
Birmingham	England	8606	10.5	3.0	14.5	7.6	3.9	1.8
Portsmouth	England	1573	8.4	3.0	9.2	7.6	2.1	1.8
Camden	England	3360	7.9	3.0	20.2	7.6	2.1	1.8
Liverpool	England	3332	7.8	3.0	26.6	7.6	3.4	1.8
Walsall	England	1845	6.2	3.0	10.3	7.6	2.0	1.8
Manchester	England	5251	5.0	3.0	19.0	7.6	2.4	1.8
Reigate and Banstead	England	872	4.9	3.0	19.2	7.6	2.0	1.8
York	England	1878	4.6	3.0	9.3	7.6	2.7	1.8
Sandwell	England	2345	4.3	3.0	12.5	7.6	2.4	1.8
Rochford	England	577	4.2	3.0	7.8	7.6	2.9	1.8
Dudley	England	2153	4.1	3.0	16.3	7.6	3.6	1.8
Derby City	England	1937	3.7	3.0	9.8	7.6	1.9	1.8
Bradford	England	4115	3.6	3.0	8.7	7.6	2.4	1.8
Cardiff	Wales	3031	3.4	3.0	12.2	7.6	2.0	1.8

14 rows returned (shown in full).

Interpretation: 14 authorities are worse than the national figure on all three measures: share below 3, backlog awaiting inspection and rating age. Birmingham stands out, with 10.5% below 3, 14.5% awaiting inspection and ratings averaging 3.9 years old. Liverpool has the largest backlog (26.6% awaiting) and Camden (20.2%) and Manchester (19.0%) are also high. These authorities are likely to be short of inspection capacity.

Q35

Question: For each business type, which of the three sub-scores is the weakest on average, and by how much?

Concept: GROUP BY + three AVGs + GREATEST / CASE to pick the weakest area

business_type	rated	avg_hygiene	avg_structural	avg_confidence	weakest_area
Takeaway/sandwich shop	50882	4.55	5.42	5.35	Structural
Restaurant/Cafe/Canteen	117408	3.47	4.36	4.16	Structural
Retailers - other	88295	2.89	3.94	4.31	Confidence in management
Pub/bar/nightclub	43450	2.90	4.38	3.56	Structural
Hotel/bed & breakfast/guest house	10703	2.57	3.29	3.28	Structural
Manufacturers/packers	7749	2.03	3.17	3.35	Confidence in management
Hospitals/Childcare/Caring Premises	35766	2.39	2.91	3.12	Confidence in management
Mobile caterer	23921	2.13	2.52	3.09	Confidence in management
Distributors/Transporters	2607	1.74	2.77	3.05	Confidence in management
Retailers - supermarkets/hypermarkets	14168	1.93	3.31	2.19	Structural
School/college/university	30740	1.93	3.08	2.36	Structural
Importers/Exporters	409	1.82	2.32	3.19	Confidence in management
Other catering premises	56624	1.93	2.11	2.97	Confidence in management
Farmers/growers	1241	1.49	2.04	2.29	Confidence in management

14 rows returned (shown in full).

Interpretation: Takeaways have the highest penalty scores in all three areas, followed by restaurants. For food service businesses (takeaways, restaurants, pubs and hotels), structural compliance is the weakest area, while for other retailers, manufacturers and care premises it is confidence in management. Inspection and advice priorities should differ by business type.

Q36

Question: Which 20 postcode areas have the highest share of businesses below 3 (1,000+ rated), and how does that relate to their share of takeaways and their average rating?

Concept: Derived column + GROUP BY with several conditional aggregates + HAVING

postcode_area	rated	avg_rating	pct_below_3	pct_takeaways
UB	2400	4.05	13.4	12.5
E	8116	4.19	10.1	13.7
IG	1923	4.23	9.4	17.5
WC	1753	4.36	8.7	8.4
B	12501	4.34	8.5	12.5
BL	2593	4.26	8.3	17.5
WN	1915	4.20	7.7	16.1
NW	4355	4.31	7.1	16.0
OL	3184	4.47	7.1	18.5
W	7014	4.47	6.9	7.2
L	5300	4.40	6.7	14.4
N	5951	4.42	6.2	11.4
HA	2895	4.40	6.0	16.2
BB	3832	4.48	5.7	17.0
M	9026	4.46	5.6	15.3
WV	2540	4.50	5.4	14.6
EN	2012	4.51	5.3	13.2
CR	2406	4.47	5.2	15.7

postcode_area	rated	avg_rating	pct_below_3	pct_takeaways
HX	1220	4.49	5.2	14.3
LU	1790	4.45	4.9	15.3

20 rows returned (shown in full).

Interpretation: The postcode areas with the most businesses below 3 are mostly in London: UB (Southall and Hayes, 13.4%), E (10.1%) and IG (9.4%), alongside Birmingham (B, 8.5%) and Greater Manchester areas such as Bolton (BL) and Wigan (WN). Takeaway share does not explain the pattern on its own: W (west London) has only 7.2% takeaways but 6.9% below 3.

Q37

Question: Among takeaways, how do chain and independent outlets compare in each nation?

Concept: CTE + window COUNT OVER + two-way GROUP BY

nation	ownership	takeaways	avg_rating	pct_rated_5	pct_below_3
England	Chain	4378	4.80	87.3	1.4
England	Independent	43161	4.29	59.3	6.2
Northern Ireland	Chain	80	4.85	88.8	0.0
Northern Ireland	Independent	1618	4.35	58.9	4.3
Wales	Chain	214	4.76	84.1	0.9
Wales	Independent	2708	4.20	51.8	5.2

6 rows returned (shown in full).

Interpretation: Independent takeaways are far weaker than chain takeaways in every nation. In England, 59.3% of independents are rated 5 and 6.2% are below 3, against 87.3% and 1.4% for chains. Welsh independents have the lowest share rated 5 (51.8%). Independent takeaways are the single clearest target group for intervention.

Q38

Question: Among businesses rated 0 or 1, which sub-score is the main problem, and does that differ by sector?

Concept: CTE + CASE comparing three columns + cross-tab by sector

sector	rated_0_or_1	pct_hygiene	pct_structural	pct_confidence
Hospitality	4055	1.5	3.0	95.5
Retail	2875	0.9	2.5	96.6
Other	305	1.6	3.6	94.8
Public sector and care	153	0.0	2.6	97.4
Production and supply	112	0.0	7.1	92.9

5 rows returned (shown in full).

Interpretation: For 95% or more of businesses rated 0 or 1 in every sector, confidence in management is the main problem. This is partly built into the scoring: the confidence score runs up to 30 rather than 25, and businesses rated 0 or 1 almost always have a high management score (Q26). Poor food safety management, rather than hygiene or premises alone, is what pushes businesses to the bottom ratings.

Q39

Question: Over the last 12 months, how many FHRS ratings were issued each month in each nation, what share were rated 5, and what is the running total for the year?

Concept: CTE + GROUP BY month and nation + window running SUM partitioned by nation

nation	rating_month	ratings	pct_rated_5	running_total
England	2025-10	1208	71.9	1208
England	2025-11	16555	74.2	17763
England	2025-12	11509	72.5	29272
England	2026-01	17346	74.1	46618
England	2026-02	18278	74.3	64896
England	2026-03	21465	74.0	86361
England	2026-04	13612	73.3	99973
England	2026-05	14493	72.7	114466
England	2026-06	16851	71.9	131317
England	2026-07	17024	70.5	148341

nation	rating_month	ratings	pct_rated_5	running_total
England	2026-08	14201	68.4	162542
England	2026-09	17921	72.1	180463
England	2026-10	7957	95.5	188420
Northern Ireland	2025-10	35	88.6	35
Northern Ireland	2025-11	638	80.9	673
Northern Ireland	2025-12	411	83.0	1084
Northern Ireland	2026-01	662	79.0	1746
Northern Ireland	2026-02	648	74.5	2394
Northern Ireland	2026-03	715	72.4	3109
Northern Ireland	2026-04	523	81.1	3632
Northern Ireland	2026-05	631	79.6	4263
Northern Ireland	2026-06	676	82.4	4939
Northern Ireland	2026-07	534	75.8	5473
Northern Ireland	2026-08	473	71.9	5946
Northern Ireland	2026-09	589	75.6	6535
Northern Ireland	2026-10	174	99.4	6709
Wales	2025-10	67	58.2	67
Wales	2025-11	1168	71.4	1235
Wales	2025-12	880	67.6	2115
Wales	2026-01	1169	69.9	3284
Wales	2026-02	1170	71.7	4454
Wales	2026-03	1441	70.0	5895
Wales	2026-04	946	70.9	6841
Wales	2026-05	1032	72.8	7873
Wales	2026-06	1211	70.5	9084
Wales	2026-07	1118	68.2	10202
Wales	2026-08	935	61.2	11137

nation	rating_month	ratings	pct_rated_5	running_total
Wales	2026-09	1184	69.7	12321
Wales	2026-10	378	98.7	12699

39 rows returned (shown in full).

Interpretation: In full months, England issues about 11,500–21,500 ratings a month, Wales about 900–1,400 and Northern Ireland 400–700. The share rated 5 is broadly stable at 70–75% in England and Wales. The last month, October 2026, shows 95–99% rated 5 because lower ratings are held back during the appeal period before publication, and October 2025 is a part month at the edge of the 12-month window.

Q40

Question: In each local authority, which business type has the lowest average rating (20+ rated), and how far is it below the LA's overall average? Show the 25 widest gaps.

Concept: CTE (two steps) + window functions (ROW_NUMBER and AVG OVER, both partitioned by LA)

la_name	nation	weakest_type	rated	type_avg_rating	la_avg_rating	gap
Fermanagh and Omagh	Northern Ireland	Takeaway/sandwich shop	100	4.11	4.79	0.68
Warrington	England	Takeaway/sandwich shop	176	3.91	4.59	0.68
Newham	England	Retailers - other	645	3.31	3.95	0.64
Cardiff	Wales	Distributors/Transporters	24	3.92	4.53	0.62
Mid Devon	England	Takeaway/sandwich shop	43	4.12	4.74	0.62
St Helens	England	Takeaway/sandwich shop	175	3.91	4.54	0.62
Bury	England	Takeaway/sandwich shop	223	3.90	4.50	0.60
Wigan	England	Takeaway/sandwich shop	340	3.61	4.21	0.60
Hillingdon	England	Takeaway/sandwich shop	261	3.71	4.31	0.60
South Kesteven	England	Takeaway/sandwich shop	99	4.15	4.75	0.59
Swansea	Wales	Takeaway/sandwich shop	218	3.95	4.53	0.58
Bath and North East Somerset	England	Takeaway/sandwich shop	112	4.02	4.59	0.57
Canterbury City	England	Takeaway/sandwich shop	111	4.09	4.66	0.57
Broadland	England	Takeaway/sandwich shop	59	4.27	4.83	0.56
Tonbridge and Malling	England	Takeaway/sandwich shop	80	4.15	4.70	0.55
Blackburn	England	Retailers - other	272	3.76	4.32	0.55
Ashfield	England	Takeaway/sandwich shop	106	4.07	4.61	0.54

la_name	nation	weakest_type	rated	type_avg_rating	la_avg_rating	gap
Knowsley	England	Retailers - supermarkets/hypermarkets	22	3.86	4.40	0.54
Bracknell Forest	England	Takeaway/sandwich shop	55	4.24	4.76	0.53
Durham	England	Takeaway/sandwich shop	521	4.10	4.63	0.53
Wokingham	England	Takeaway/sandwich shop	52	4.19	4.72	0.53
Lancaster City	England	Takeaway/sandwich shop	114	4.15	4.67	0.52
Reading	England	Retailers - other	206	3.86	4.37	0.52
West Northamptonshire	England	Takeaway/sandwich shop	227	4.11	4.63	0.52
Rhondda Cynon Taf	Wales	Takeaway/sandwich shop	259	3.98	4.51	0.52

25 rows returned (shown in full).

Interpretation: In 20 of the 25 authorities with the largest gaps, takeaways are the weakest business type, averaging 0.5–0.7 points below the local average. The exceptions are mostly other retailers, notably in Newham (3.31), Blackburn and Reading. Takeaways are the most consistent local weak spot.

7. Summary of Findings

Food hygiene compliance is high overall, but the problems are concentrated in independent takeaways and retailers, in inner-city areas, and in authorities that are falling behind on inspections.

Section	Key finding
KPI	97.0% of rated businesses are broadly compliant and 78.5% are rated 5, leaving 14,876 below 3 (Q2, Q3). 42,301 FHRS businesses are awaiting a first inspection (Q5), and a quarter of ratings are over two years old (Q7).
Univariate	Ratings are heavily skewed towards 5 (Q11). Structural and management scores are perfect less often than hygiene (Q13–Q15), and some ratings date back to 1998 (Q17).

Section	Key finding
Bivariate	Takeaways (5.7% below 3) and other retailers (5.2%) are the weakest types (Q21). Northern Ireland outperforms England and Wales (Q23), and chain outlets outperform independents (Q30). Recent ratings are lower because weaker businesses are inspected more often (Q24).
Multivariate	Independent takeaways are the clearest problem group, with only 52–59% rated 5 (Q37). The worst authorities are mostly in London and the West Midlands (Q33), and 14 authorities, led by Birmingham, are behind on compliance, backlog and rating age at once (Q34). Poor confidence in management is behind almost every 0 or 1 rating (Q38).

For an environmental health manager, the priorities are independent takeaways and retailers, management training rather than premises alone, and clearing the inspection backlog in the authorities identified in Q34.

The main limitations are that Scotland's FHIS cannot be compared directly with FHRS, the most recent month understates low ratings because of the appeal period (Q39), and the file holds only each business's current rating, so changes over time for individual businesses cannot be tracked.